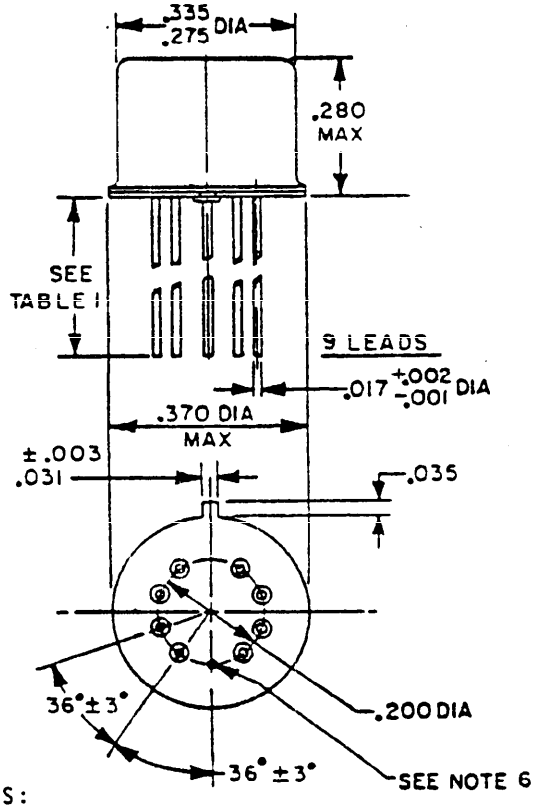
MILITARY SPECIFICATION SHEET

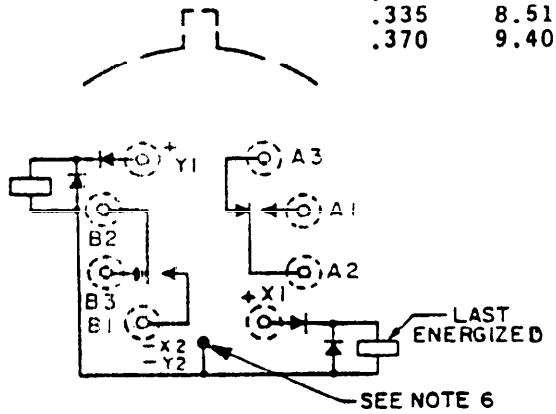
RELAYS, ELECTROMAGNETIC, ESTABLISHED RELIABILITY, DPDT, LOW LEVEL TO 1.0 AMPERE (LATCHING) WITH INTERNAL DIODES FOR COIL TRANSIENT SUPPRESSION AND POLARITY REVERSAL PROTECTION, INCLUDING INSPECTIONS OF APPENDIXES A AND B

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-R-39016.



Inches	mm
.001	0.03
.002	0.05
.003	0.08
.017	0.43
.031	0.79
.035	0.89
.200	5.08
.275	6.98
.280	7.11
.335	8.51
.370	9.40



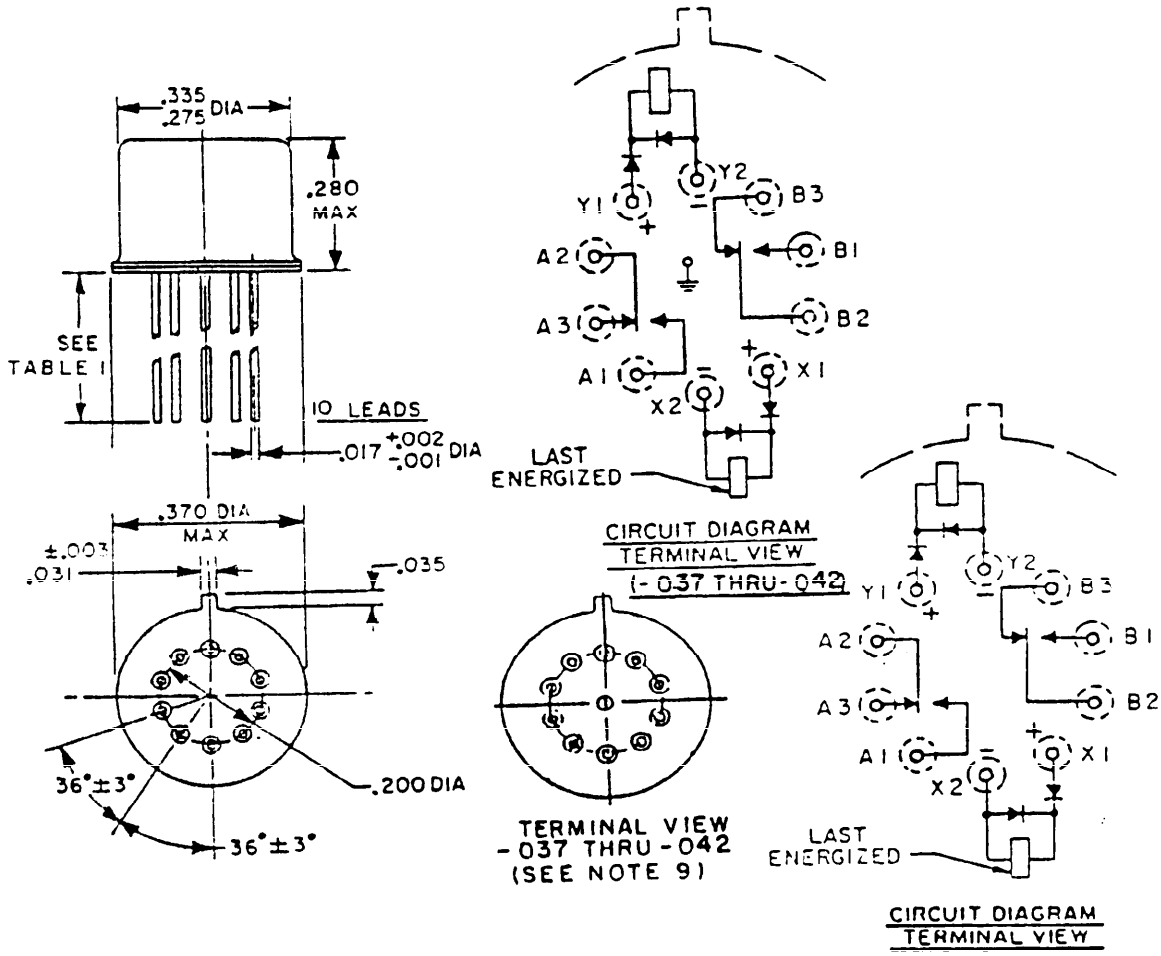
CIRCUIT DIAGRAM
TERMINAL VIEW

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
4. Terminal numbers shown above for reference only. Numbers do not appear on the relay.
5. Relays shall have a plus (+) sign placed on the circuit diagram as shown.
6. All leads shall be electrically insulated from the case, except for lead terminal -X2 -Y2, which is grounded to the case.
7. Coil symbol optional in accordance with MIL-STD-1285.
8. Circuit diagram shown on part is the terminal view.

FIGURE 1. Dimensions and configuration (round).

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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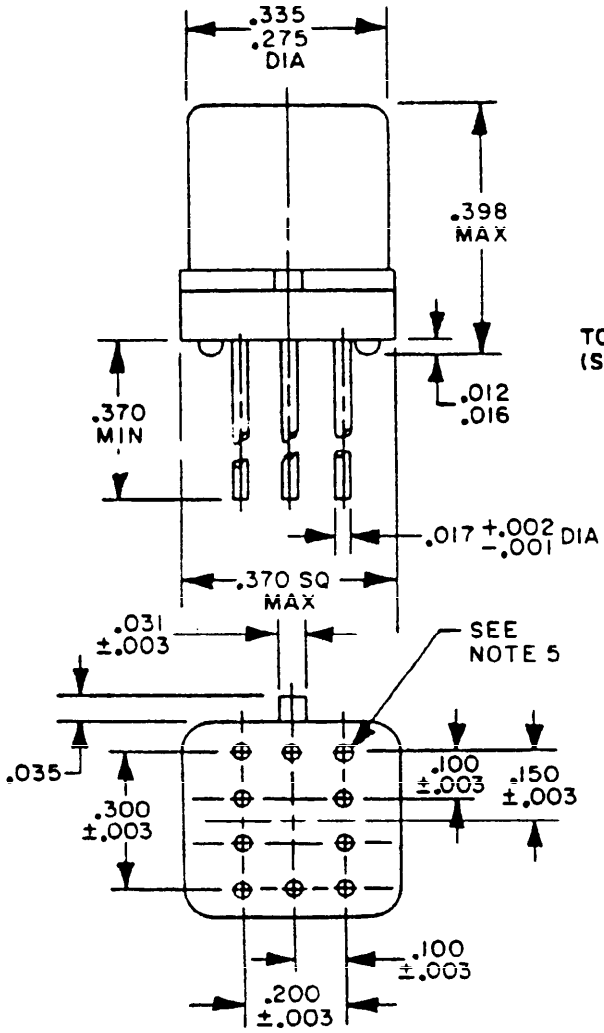


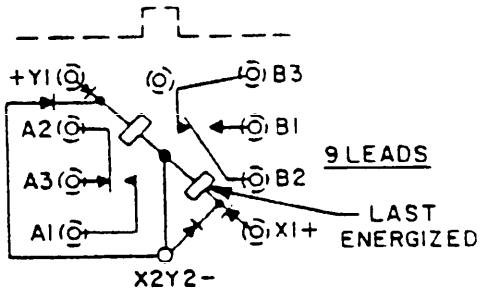
Inches	mm	Inches	mm	Inches	mm
.001	0.03	.031	0.79	.275	6.98
.002	0.05	.035	0.89	.280	7.11
.003	0.08	.200	5.08	.335	8.51
.017	0.43			.370	9.40

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
4. Terminal numbers shown above for reference only. Numbers do not appear on the relay.
5. Relays shall have a plus (+) sign placed on the circuit diagram as shown.
6. All leads shall be electrically insulated from the case.
7. Coil symbol optional in accordance with MIL-STD-1285.
8. Circuit diagram shown on part is the terminal view.
9. M39016/630-037 through M39016/630-042 shall be supplied with a case grounding pin welded to the relay header as shown.

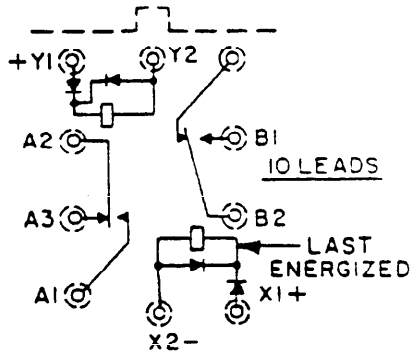
FIGURE 2. Dimensions and configuration.

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CIRCUIT DIAGRAM 1
TERMINAL VIEW APPLICABLE
TO DASH NUMBERS -025 THRU -030
(SEE NOTE 9)



CIRCUIT DIAGRAM 2
TERMINAL VIEW APPLICABLE
TO DASH NUMBERS -031 THRU -036

Inches	mm	Inches	mm
.001	0.03	.100	2.54
.002	0.05	.150	3.81
.003	0.08	.200	5.08
.012	0.30	.275	6.99
.016	0.41	.300	7.62
.017	0.43	.335	8.51
.031	0.79	.370	9.40
.035	0.89	.398	10.11

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
4. Spreader pads shall be certified to MIL-M-38527, M38527/5-003 or M38527/5-013.
5. Dimensions and tolerances shown for the bottom view of the spreader pad are for the center to center locations of the holes in the spreader pad.
6. Shape optional within the envelope dimension.
7. Terminal numbers shown above for reference only. Numbers do not appear on the relay.
8. Relays shall have a plus (+) sign placed on the circuit diagram as shown.
9. All leads shall be electrically insulated from the case, except for lead terminal -X2 -Y2, which is grounded to the case.
10. Coil symbol optional in accordance with MIL-STD-1285.
11. Circuit diagram shown on part is the terminal view.

FIGURE 3. Dimensions and configuration relay with spreader pad attached.

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REQUIREMENTS:

Cleaning and small particle inspection: Applicable.

CONTACT DATA:

Load ratings:

High level (relay case grounded):

Resistive:

1.0 ampere at 28 V dc.

500 milliamperes at 115 V ac 400 Hz case not grounded.

250 milliamperes at 115 V ac 60 Hz case not grounded.

100 milliamperes at 115 V ac 60 and 400 Hz case grounded.

Inductive load: 0.2 ampere at 28 V dc with 0.32 henry inductance.

Lamp: 0.10 ampere at 28 V dc.

Low level: 10 to 50 µA at 10 to 50 mV dc or peak ac.

Intermediate current: Applicable.

Contact resistance or voltage drop:

Initial: 0.125 ohm maximum (0.150 ohm maximum with spreader pads attached).

High level:

During life: Not more than 5 percent of open circuit voltage.

After life: 0.225 ohm maximum (0.250 ohm maximum with spreader pad attached).

Low level:

During life: 33 ohms maximum.

After life: 0.175 ohm maximum (0.200 ohm maximum with spreader pad attached).

Intermediate current:

During: 1 ohm maximum.

After: 0.225 ohm maximum (0.250 ohm maximum with spreader pad attached).

Contact bounce: 2 milliseconds (ms) maximum (applicable for after high level rated load life and intermediate current tests).

Contact stabilization time: 2.5 milliseconds maximum.

Overload (high level only): Two times rated current.

Neutral screen: Applicable.

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COIL DATA: See table I.

Operate time: 2.0 ms maximum over temperature range with rated coil voltage.

Release time: Not applicable.

ELECTRICAL DATA: 1/ 2/

Insulation resistance: 10,000 megohms minimum at 500 V dc, except the resistance between coil and case at high temperature shall be 1,000 megohms minimum.

Dielectric withstanding voltage:

	Sea level V rms (60 Hz)	Altitude V rms (60 Hz)
Between case, frame, or enclosure and all contacts both in the energized and deenergized positions	500	125 All terminals to case
Between case, frame, or enclosure and coils	500	
Between all contacts and coils	500	
Between open contacts in the energized and deenergized positions	500	
Between contact poles	500	
Between coils (applicable to 10 and 11 lead relays)	500	

DIODE CHARACTERISTICS: 3/

Maximum transient voltage: 1 volt.

Breakdown voltage: 100 V dc minimum at 10 microamperes (μA).

Maximum leakage current: 1 μA at 50 V dc.

Coil transient suppression: Applicable.

Semiconductor in process screening: Applicable, visual inspection of semiconductors shall be accordance with MIL-STD-750 method 2073.

1/ Insulation resistance and dielectric withstanding voltage tests must always precede all other specified electrical measurements. Connect all coil terminals together to avoid damage to diodes.

2/ Dielectric withstanding voltage and insulation resistance are not applicable between coils and case or from coil to coil on figure 1 relays.

3/ In all tables relating to qualification and group A testing, delete coil resistance and substitute the following test:

a. Diode breakdown and block integrity:
With applicable voltage applied to the relay coil circuit in the reverse direction, monitor leakage current with dc microammeter or oscilloscope or qualifying activity approved test equipment. Leakage current shall not exceed the specified value.

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ENVIRONMENTAL DATA:

Temperature range: -65°C to +125°C.

Vibration (sinusoidal): MIL-STD-202, method 204. Contact chatter shall not exceed 10 microseconds maximum for closed contacts, and 1 microsecond maximum closure for open contacts.

Vibration (random): MIL-STD-202, method 214, test condition IG. Contact chatter shall not exceed 10 microseconds maximum for closed contacts, and 1 microsecond maximum closure for open contact. Applicable to qualification, group C, and group A inspection testing.

Shock (specified pulse): MIL-STD-202, method 213, test condition B (75 G's). Contact chatter shall not exceed 10 microseconds maximum for closed contacts, and 1 microsecond maximum closure for open contacts.

Magnetic interference: Applicable.

Resistance to soldering heat: Applicable.

Acceleration: Applicable.

Salt atmosphere (corrosion): In accordance with method 1041, MIL-STD-750.

Particle impact noise detection: Applicable.

PHYSICAL DATA:

Terminal strength (method 211, MIL-STD-202):

Pull test: Test condition A, 1 pound pull.

Bend test: Test condition C, 1/2 pound load.

Twist test: Test condition D as specified in MIL-R-39016.

Solderability: Applicable.

Dimensions and configurations: See figures 1, 2, and 3.

Weight: 2.84 grams (0.10 ounce) maximum, 3.09 grams (0.109 ounce) maximum with spreader pad attached.

Seal: Hermetic, adjunct sealant shall not be used.

Minimum marking: Military PIN, "J" with the date code (example J8830), circuit diagram, and manufacturer's name or source code.

LIFE TEST REQUIREMENTS:

High level: 100,000 cycles per relay.

Low level: 100,000 cycles plus 900,000 cycles mechanical life.

Part or Identifying Number (PIN): M39016/630- (dash number from table I and suffix letter designating failure rate level).

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TABLE I. Dash numbers and characteristics. 1/ 2/

Dash numbers 3/				.500 Min with ground pin 6/	Number of leads	Coil voltge V dc 4/		At 25°C				Over temp range
Lead length .187 ±.010	Lead length .500 min	Spreader pad (fig. 3) 5/				Rated	Max	Coil resist-ohms (Ref only) 7/	Coil circuit current 4/ (mA) Max	Coil circuit current 4/ (mA) Min	Specified pickup (latch/ reset) value (volt-age) (V dc)	Specified pickup (latch/ reset) value (volt-age) (V dc)
001	013	025	Circuit	---	9	6.0	8.0	97	63.0	46.9	4.1	5.5
002	014	026	diagram	---	9	9.0	12	280	33.7	26.0	6.3	7.8
003	015	027	1	---	9	12	16	500	25.5	20.0	8.0	10
004	016	028		---	9	18	24	1,130	17.2	13.7	11.6	14.5
005	017	029		---	9	26.5	32	2,000	14.4	11.6	15.4	19
006	018	030		---	9	5.0	6.0	48	104.2	75.8	3.5	4.5
007	019	031	Circuit	---	10	5.0	6.0	48	104.2	75.8	3.5	4.5
008	020	032	diagram	---	10	6.0	8.0	97	63.0	46.9	4.1	5.5
009	021	033	2	---	10	9.0	12	280	33.7	26.0	6.3	7.8
010	022	034		---	10	12	16	500	25.5	20.0	8.0	10
011	023	035		---	10	18	24	1,130	17.2	13.7	11.6	14.5
012	024	036		---	10	26.5	32	2,000	14.4	11.6	15.4	19
---	---	---		037	11	12	16	48	104.2	75.8	3.5	4.5
---	---	---		038	11	5.0	6.0	97	63.0	46.9	4.1	5.5
---	---	---		039	11	6.0	8.0	280	33.7	26.0	6.3	7.8
---	---	---		040	11	9.0	12	500	25.5	20.0	8.0	10
---	---	---		041	11	18	24	1,130	17.2	13.7	11.6	14.5
---	---	---		042	11	26.5	32	2,000	14.4	11.0	15.4	19

1/ Each relay possesses high level and low level capabilities. However, relays previously tested or used above 10 mA resistive at 6 V dc maximum or peak ac open circuits not recommended for subsequent use in low level applications.

2/ Warning: When latching relays are installed in equipment, the latch and reset coils should not be pulsed simultaneously. Coils should not be pulsed with less than the nominal coil voltage and the pulse width should be a minimum of three times the specified operate time of the relay. If these conditions are not followed, it is possible for the relay to be in the magnetically neutral position.

3/ The suffix letter M, P, or R to designate the applicable failure rate level shall be added to the applicable listed dash number. Failure rate level (percent per 10,000 cycles): M, 1.0; P, 0.1; R, 0.01. Example, 025M - - - - 042R.

4/ Caution: The use of any coil voltage less than the rated coil voltage will compromise the operation of the relay.

5/ Relays supplied with spreader pads (-025 through -036) shall have the pads rigidly attached.

6/ Relays are supplied with a case grounding pin welded to the header (see figure 2).

7/ Coil resistance not directly measurable at relay terminals. When rated voltage is applied to coil terminals, the coil circuit must be within the limits shown. Measure at 25°C at nominal voltage for 5 seconds maximum.

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QUALIFICATION INSPECTION:

Qualification inspection and sample size: See table II.

Failure rate qualification: The manufacturer shall qualify his products to failure rate level "M" initially.

TABLE II. Qualification inspection and sample size. 1/

Single submission	Group submission	
33 units plus 1 open unit for level M at C = 0 2/ Qualification inspection as applicable	M39016/630-017	33 units plus 1 open unit for level M at C = 0 2/ Qualification inspection as applicable
	M39016/630-013 M39016/630-014 M39016/630-015 M39016/630-016 M39016/630-018 M39016/630-019 M39016/630-020 M39016/630-021 M39016/630-022 M39016/630-023 M39016/630-024	2 units each PIN Qualification inspection, group I, group II
	M39016/630-042	1 unit terminal strength and solderability
	M39016/630-029	2 units each PIN 3/

1/ For retention of qualification or extension of qualification to lower failure rate levels, all life test data accumulated on MIL-R-39016/30 may be used in addition to MIL-R-39016/629. Prior to performance of retention of qualification testing, the relay manufacturer shall preselect the sampling plan.

2/ The number of units required for qualification testing shall be increased as required in group V, table I, MIL-R-39016, if the contractor elects to test the number of units permitting one or more failures. Prior to performance of qualification inspection, the relay manufacturer shall preselect the sampling plan.

3/ Qualification inspection of relays supplied with spreader pads shall be tested as specified in the qualification inspection table of MIL-R-39016 in the order shown below:

Before installation of the pad:

Vibration (sinusoidal) test duration shall be 10 minutes, vibration (random), particle impact noise detection, screening.

After installation of the pad:

Insulation resistance, dielectric withstanding voltage, static contact resistance, specified pickup, hold, and dropout values (voltages), coil resistance, operate and release time, contact dynamic characteristics, coil transient suppression (when specified), solderability, seal, visual and mechanical inspection (external).

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Group A inspection: Group A inspection for relays supplied with spreader pads shall be as specified below:

Before installation of the pad: Perform subgroups 1 and 2 of the group A inspection table.

After installation of the pad: Perform subgroups 3 and 4 of the group A inspection table.

Qualification inspection (reduced testing) and sample size: If the relays produced for MIL-R-39016/630 are similar in construction and design, except for the additional inspections specified herein, to the relays produced for MIL-R-39016/30, then reduced testing may be performed concurrent with or subsequent to successful qualification of MIL-R-39016/30 at the "M" failure rate level. See table III.

TABLE III. Qualification inspection (reduced testing).

Examination of test
2 units each coil voltage, group A inspection table. 1 unsealed sample unit for internal examination.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AL, AR, AT, MI
Navy - OS, SH,
Air Force - 99
DLA - ES

User activities:
Army - AV, ME
Navy - AS, MC
Air Force - 11, 19

Preparing activity:
Navy - EC

Agent:
DLA - ES

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